TUGAS PRATIKUM
JARINGAN KOMPUTER 4

ANDI MUH ARIEF ALFAIZI ILHAM (103072400082)

PROGRAM STUDI INFORMATIKA
FAKULTAS INFORMATIKA
TELKOM UNIVERSITY
SURABAYA

4.2. nslookup

- .

```
C:\Users\ASUS ROG STRIX>nslookup facebook.com
Server:  UnKnown
Address:  10.243.33.46

Non-authoritative answer:
Name:    facebook.com
Addresses:  2a03:2880:f14d:181:face:b00c:0:25de
          31.13.95.35


C:\Users\ASUS ROG STRIX>nslookup -type=NS facebook.com
Server:  UnKnown
Address:  10.243.33.46

Non-authoritative answer:
facebook.com    nameserver = a.ns.facebook.com
facebook.com    nameserver = c.ns.facebook.com
facebook.com    nameserver = b.ns.facebook.com
facebook.com    nameserver = d.ns.facebook.com
```

- .

```
C:\Users\ASUS ROG STRIX>nslookup www.facebook.com 8.8.8.8
Server:  dns.google
Address:  8.8.8.8

Non-authoritative answer:
Name:    star-mini.c10r.facebook.com
Addresses:  2a03:2880:f14d:181:face:b00c:0:25de
          31.13.95.35
Aliases:  www.facebook.com
```

- Jalankan nslookup untuk mendapatkan alamat IP dari server web di Asia. Berapa alamat IP server tersebut?
  alamat IP dari server web National University of Singapore adalah **45.60.35.225**

```
C:\Users\ASUS ROG STRIX>nslookup www.nus.edu.sg
Server:  UnKnown
Address:  10.243.33.46

Non-authoritative answer:
Name:    mgnzsqc.x.incapdns.net
Addresses:  64:ff9b::2d3c:23e1
          45.60.35.225
Aliases:  www.nus.edu.sg


C:\Users\ASUS ROG STRIX>nslookup -type=NS cam.ac.uk
Server:  UnKnown
Address:  10.243.33.46
```

- Jalankan nslookup agar dapat mengetahui server DNS otoritatif untuk universitas di Eropa server DNS otoritatif untuk University of Cambridge antara lain adalah **dns0.eng.cam.ac.uk, auth0.dns.cam.ac.uk, dll**.

```
C:\Users\ASUS ROG STRIX>nslookup -type=NS cam.ac.uk
Server:  UnKnown
Address:  10.243.33.46

Non-authoritative answer:
cam.ac.uk       nameserver = dns0.eng.cam.ac.uk
cam.ac.uk       nameserver = ns1.mythic-beasts.com
cam.ac.uk       nameserver = auth0.dns.cam.ac.uk
cam.ac.uk       nameserver = auth1.dns.cam.ac.uk
cam.ac.uk       nameserver = ns2.ic.ac.uk
cam.ac.uk       nameserver = dns0.cl.cam.ac.uk
cam.ac.uk       nameserver = ns3.mythic-beasts.com
```

- Jalankan nslookup untuk mencari tahu informasi mengenai server email dari Yahoo! Mail melalui salah satu server yang didapatkan di pertanyaan nomor 2. Apa alamat IP-nya?

```
C:\Users\ASUS ROG STRIX>nslookup -type=MX yahoo.com auth0.dns.cam.ac.uk
Server:  auth0.dns.cam.ac.uk
Address:  131.111.8.37

*** auth0.dns.cam.ac.uk can't find yahoo.com: Query refused

C:\Users\ASUS ROG STRIX>nslookup -type=MX yahoo.com 8.8.8.8
Server:  dns.google
Address:  8.8.8.8

Non-authoritative answer:
yahoo.com       MX preference = 1, mail exchanger = mta6.am0.yahoodns.net
yahoo.com       MX preference = 1, mail exchanger = mta7.am0.yahoodns.net
yahoo.com       MX preference = 1, mail exchanger = mta5.am0.yahoodns.net
```

4.3. Ipconfig

- ipconfig /all

```
Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) Dual Band Wireless-AC 7265
   Physical Address. . . . . . . . . : 48-45-20-FA-27-B0
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   IPv6 Address. . . . . . . . . . . : 2404:c0:317f:694:fb76:535b:80be:8d5b(Preferred)
   Temporary IPv6 Address. . . . . . : 2404:c0:317f:694:9d55:b7fd:d37a:7f78(Preferred)
   Link-local IPv6 Address . . . . . : fe80::d64:472d:9285:d7cc%6(Preferred)
   IPv4 Address. . . . . . . . . . . : 10.243.33.126(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Tuesday, 31 March 2026 12:56:02
   Lease Expires . . . . . . . . . . : Tuesday, 31 March 2026 13:56:01
   Default Gateway . . . . . . . . . : fe80::94fb:67ff:fe6f:765c%6
                                       10.243.33.46
   DHCP Server . . . . . . . . . . . : 10.243.33.46
   DHCPv6 IAID . . . . . . . . . . . : 105399584
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2E-4C-01-72-60-45-CB-BD-63-AF
   DNS Servers . . . . . . . . . . . : 10.243.33.46
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

- , ipconfig /displaydns

```
www.google.com
----------------------------------------
Record Name . . . . . : www.google.com
Record Type . . . . . : 5
Time To Live  . . . . : 225
Data Length . . . . . : 8
Section . . . . . . . : Answer
CNAME Record  . . . . : forcesafesearch.google.com


Record Name . . . . . : forcesafesearch.google.com
Record Type . . . . . : 1
Time To Live  . . . . : 225
Data Length . . . . . : 4
Section . . . . . . . : Answer
A (Host) Record . . . : 216.239.38.120
```

- ipconfig /flushdns

```
C:\Users\ASUS ROG STRIX>ipconfig /flushdns

Windows IP Configuration

Successfully flushed the DNS Resolver Cache.
```

4.4. **Tracing DNS dengan Wireshark**

- . Cari pesan permintaan DNS dan balasannya. Apakah pesan tersebut dikirimkan melalui UDP atau TCP?

  UDP

910	17.767327400	10.243.33.126	10.243.33.46	DNS	72	Standard query 0xc853 AAAA www.ietf.org
911	17.767541200	10.243.33.126	10.243.33.46	DNS	72	Standard query 0x2885 A www.ietf.org

- Apa port tujuan pada pesan permintaan DNS? Apa port sumber pada pesan balasannya?

  (*Destination Port* / Dst Port) pada pesan **permintaan** DNS adalah **53**.

  Sedangkan port sumber (*Source Port* / Src Port) pada pesan **balasan** DNS juga adalah **53**.

```
▾ User Datagram Protocol, Src Port: 52693, Dst Port: 53
```

```
▾ User Datagram Protocol, Src Port: 53, Dst Port: 52693
```

- Pada pesan permintaan DNS, apa alamat IP tujuannya? Apa alamat IP server DNS lokal anda (gunakan ipconfig untuk mencari tahu)? Apakah kedua alamat IP tersebut sama?

  Internet Protocol Version 4 pada Wireshark ip nya 10.243.33.46.

  ipconfig /all ip nya 10.243.33.46.

  jadi kedua alamat IP tersebut SAMA

```
▾ Internet Protocol Version 4, Src: 10.243.33.126, Dst: 10.243.33.46
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  ▸ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 58
    Identification: 0x4822 (18466)
  ▸ 000. .... = Flags: 0x0
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 128
    Protocol: UDP (17)
    Header Checksum: 0x99ff [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 10.243.33.126
    Destination Address: 10.243.33.46
    [Stream index: 1]
```

```
Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) Dual Band Wireless-AC 7265
   Physical Address. . . . . . . . . : 48-45-20-FA-27-B0
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   IPv6 Address. . . . . . . . . . . : 2404:c0:317f:694:fb76:535b:80be:8d5b(Preferred)
   Temporary IPv6 Address. . . . . . : 2404:c0:317f:694:9d55:b7fd:d37a:7f78(Preferred)
   Link-local IPv6 Address . . . . . : fe80::d64:472d:9285:d7cc%6(Preferred)
   IPv4 Address. . . . . . . . . . . : 10.243.33.126(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Tuesday, 31 March 2026 12:56:03
   Lease Expires . . . . . . . . . . : Tuesday, 31 March 2026 14:26:00
   Default Gateway . . . . . . . . . : fe80::94fb:67ff:fe6f:765c%6
                                       10.243.33.46
   DHCP Server . . . . . . . . . . . : 10.243.33.46
   DHCPv6 IAID . . . . . . . . . . . : 105399584
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2E-4C-01-72-60-45-CB-BD-63-AF
   DNS Servers . . . . . . . . . . . : 10.243.33.46
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

- Periksa pesan permintaan DNS. Apa “jenis” atau ”type” dari pesan tersebut? Apakah pesan permintaan tersebut mengandung ”jawaban” atau ”answers”?

  type dari pesan permintaan DNS tersebut adalah **Type AAAA**

```
▾ Domain Name System (query)
    Transaction ID: 0xc853
  ▸ Flags: 0x0100 Standard query
    Questions: 1
    Answer RRs: 0
    Authority RRs: 0
    Additional RRs: 0
  ▸ Queries
    [Response In: 951]
```

- Periksa pesan balasan DNS. Berapa banyak ”jawaban” atau ”answers” yang terdapat di dalamnya? Apa saja isi yang terkandung dalam setiap jawaban tersebut?
  Ada 2 yaitu
  2606:4700::6810:2c63
  2606:4700::6810:2d63

```
Domain Name System (response)
    Transaction ID: 0xc853
  ▸ Flags: 0x8180 Standard query response, No error
    Questions: 1
    Answer RRs: 2
    Authority RRs: 0
    Additional RRs: 0
  ▸ Queries
  ▾ Answers
    ▸ www.ietf.org: type AAAA, class IN, addr 2606:4700::6810:2c63
    ▸ www.ietf.org: type AAAA, class IN, addr 2606:4700::6810:2d63
    [Request In: 910]
    [Time: 148.023300 milliseconds]
```

- . Perhatikan paket TCP SYN yang selanjutnya dikirimkan oleh host Anda. Apakah alamat IP pada paket tersebut sesuai dengan alamat IP yang tertera pada pesan balasan DNS?
  YA, Sesuai. Berdasarkan pengamatan pada paket nomor **985**, *host* mengirimkan paket **TCP [SYN]** ke alamat tujuan **2606:4700::6810:2d63**.

```
985 17.957238500  2404:c0:317f:694:9d… 2606:4700::6810:2d63 TCP   86 57426 → 443 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 WS=256 SACK_PERM
996 18.486760600  2606:4700::6810:2d63 2404:c0:317f:694:9d… TCP   86 443 → 57426 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1400 SACK_PERM WS=512
```

- . Halaman web yang sebelumnya anda akses (http://www.ietf.org) memuat beberapa gambar. Apakah host Anda perlu mengirimkan pesan permintaan DNS baru setiap kali ingin mengakses suatu gambar?
  TIDAK. Setelah *host* melakukan permintaan DNS pertama kali untuk domain www.ietf.org, *host* akan mendapatkan alamat IP server tersebut dan menyimpannya di dalam **DNS Cache** seperti yang dibuktikan pada percobaan ipconfig /displaydns sebelumnya.

nslookup www.mit.edu

```
C:\Users\ASUS ROG STRIX>nslookup www.mit.edu
Server:  UnKnown
Address:  10.243.33.46

Non-authoritative answer:
Name:    e9566.dscb.akamaiedge.net
Addresses:  2404:c0:3001:192::255e
          2404:c0:3001:18a::255e
          23.217.163.122
Aliases:  www.mit.edu
          www.mit.edu.edgekey.net
```

- Apa port tujuan pada pesan permintaan DNS? Apa port sumber pada pesan balasan DNS? port tujuan (*Destination Port*) pada pesan **permintaan** DNS adalah **53**.
  Sedangkan port sumber (*Source Port*) pada pesan **balasan** DNS juga adalah **53**.

```
User Datagram Protocol, Src Port: 61769, Dst Port: 53
User Datagram Protocol, Src Port: 53, Dst Port: 61769
```

- Ke alamat IP manakah pesan permintaan DNS dikirimkan? Apakah alamat IP tersebut merupakan default alamat IP server DNS lokal Anda?
  YA, alamat IP tersebut sama dengan *default* alamat IP server DNS lokal yang tercatat pada konfigurasi jaringan komputer saat pengujian menggunakan perintah ipconfig.

```
▾ Internet Protocol Version 4, Src: 10.243.33.126, Dst: 10.243.33.46
```

```
Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2404:c0:317f:694:fb76:535b:80be:8d5b
   Temporary IPv6 Address. . . . . . : 2404:c0:317f:694:a8b4:5c1f:e40:ea45
   Link-local IPv6 Address . . . . . : fe80::d64:472d:9285:d7cc%6
   IPv4 Address. . . . . . . . . . . : 10.243.33.126
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::805d:a3ff:fe62:d75a%6
                                       10.243.33.46
```

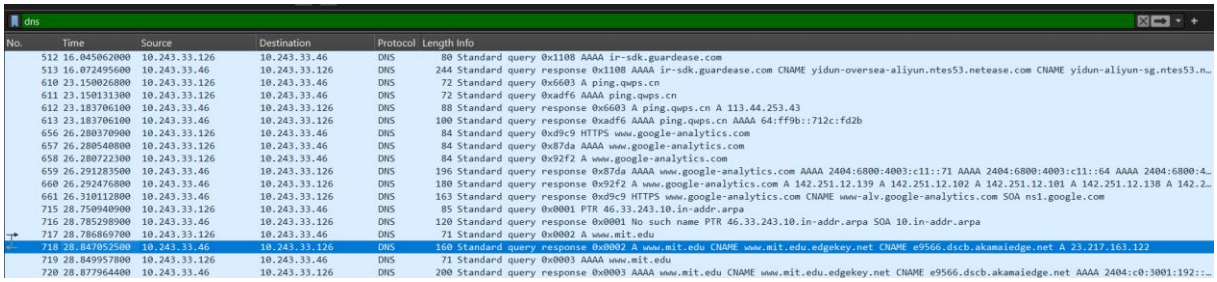
- Periksa pesan permintaan DNS. Apa ”jenis” atau ”type” dari pesan tersebut? Apakah pesan tersebut mengandung ”jawaban” atau ”answers”?
  Pesan permintaan tersebut **tidak mengandung jawaban**. Hal ini dibuktikan dengan indikator **Answer RRs: 0** pada detail paket.

```
▾ Domain Name System (query)
    Transaction ID: 0x0002
  ▸ Flags: 0x0100 Standard query
    Questions: 1
    Answer RRs: 0
    Authority RRs: 0
    Additional RRs: 0
  ▸ Queries
    [Response In: 718]
```

- Periksa pesan balasan DNS. Berapa banyak ”jawaban” atau “answers” yang terdapat di dalamnya. Apa saja isi yang terkandung dalam setiap jawaban tersebut?
  Ada 3 Yaitu:
  www.mit.edu: type CNAME, class IN, cname www.mit.edu.edgekey.net
  www.mit.edu.edgekey.net: type CNAME, class IN, cname e9566.dscb.akamaiedge.net
  e9566.dscb.akamaiedge.net: type A, class IN, addr 23.217.163.122

```
▾ Domain Name System (response)
    Transaction ID: 0x0002
  ▸ Flags: 0x8180 Standard query response, No error
    Questions: 1
    Answer RRs: 3
    Authority RRs: 0
    Additional RRs: 0
  ▸ Queries
  ▾ Answers
    ▸ www.mit.edu: type CNAME, class IN, cname www.mit.edu.edgekey.net
    ▸ www.mit.edu.edgekey.net: type CNAME, class IN, cname e9566.dscb.akamaiedge.net
    ▸ e9566.dscb.akamaiedge.net: type A, class IN, addr 23.217.163.122
    [Request In: 717]
    [Time: 60.182800 milliseconds]
```

- Sertakan hasil tangkapan layar.



dns

No.	Time	Source	Destination	Protocol	Length	Info
512	16.045062000	10.243.33.126	10.243.33.46	DNS	80	Standard query 0x1108 AAAA ir-sdk.guardease.com
513	16.072495600	10.243.33.46	10.243.33.126	DNS	244	Standard query response 0x1108 AAAA ir-sdk.guardease.com CNAME yidun-oversea-aliyun.ntes53.netease.com CNAME yidun-aliyun-sg.ntes53.n…
610	23.150026800	10.243.33.126	10.243.33.46	DNS	72	Standard query 0x6603 A ping.qwps.cn
611	23.150131300	10.243.33.126	10.243.33.46	DNS	72	Standard query 0xadf6 AAAA ping.qwps.cn
612	23.183706100	10.243.33.46	10.243.33.126	DNS	88	Standard query response 0x6603 A ping.qwps.cn A 113.44.253.43
613	23.183706100	10.243.33.46	10.243.33.126	DNS	100	Standard query response 0xadf6 AAAA ping.qwps.cn AAAA 64:ff9b::712c:fd2b
656	26.280370900	10.243.33.126	10.243.33.46	DNS	84	Standard query 0xd9c9 HTTPS www.google-analytics.com
657	26.280540800	10.243.33.126	10.243.33.46	DNS	84	Standard query 0x87da AAAA www.google-analytics.com
658	26.280722300	10.243.33.126	10.243.33.46	DNS	84	Standard query 0x92f2 A www.google-analytics.com
659	26.291283500	10.243.33.46	10.243.33.126	DNS	196	Standard query response 0x87da AAAA www.google-analytics.com AAAA 2404:6800:4003:c11::71 AAAA 2404:6800:4003:c11::64 AAAA 2404:6800:4…
660	26.292476800	10.243.33.46	10.243.33.126	DNS	180	Standard query response 0x92f2 A www.google-analytics.com A 142.251.12.139 A 142.251.12.102 A 142.251.12.101 A 142.251.12.138 A 142.2…
661	26.310112800	10.243.33.46	10.243.33.126	DNS	163	Standard query response 0xd9c9 HTTPS www.google-analytics.com CNAME www-alv.google-analytics.com SOA ns1.google.com
715	28.750940900	10.243.33.126	10.243.33.46	DNS	85	Standard query 0x0001 PTR 46.33.243.10.in-addr.arpa
716	28.785298900	10.243.33.46	10.243.33.126	DNS	120	Standard query response 0x0001 No such name PTR 46.33.243.10.in-addr.arpa SOA 10.in-addr.arpa
717	28.786869700	10.243.33.126	10.243.33.46	DNS	71	Standard query 0x0002 A www.mit.edu
718	28.847052500	10.243.33.46	10.243.33.126	DNS	160	Standard query response 0x0002 A www.mit.edu CNAME www.mit.edu.edgekey.net CNAME e9566.dscb.akamaiedge.net A 23.217.163.122
719	28.849957800	10.243.33.126	10.243.33.46	DNS	71	Standard query 0x0003 AAAA www.mit.edu
720	28.877964400	10.243.33.46	10.243.33.126	DNS	200	Standard query response 0x0003 AAAA www.mit.edu CNAME www.mit.edu.edgekey.net CNAME e9566.dscb.akamaiedge.net AAAA 2404:c0:3001:192::…

nslookup –type=NS mit.edu

```
C:\Users\ASUS ROG STRIX>nslookup -type=NS mit.edu
Server:  UnKnown
Address:  10.243.33.46

Non-authoritative answer:
mit.edu nameserver = ns1-37.akam.net
mit.edu nameserver = use5.akam.net
mit.edu nameserver = use2.akam.net
mit.edu nameserver = asia1.akam.net
mit.edu nameserver = eur5.akam.net
mit.edu nameserver = asia2.akam.net
mit.edu nameserver = usw2.akam.net
mit.edu nameserver = ns1-173.akam.net

C:\Users\ASUS ROG STRIX>
```

- Ke alamat IP manakah pesan permintaan DNS dikirimkan? Apakah alamat IP tersebut merupakan default alamat IP server DNS lokal Anda?
  YA, alamat IP tersebut sama dengan *default* alamat IP server DNS lokal yang tercatat pada konfigurasi jaringan komputer saat pengujian menggunakan perintah ipconfig.

```
▾ Internet Protocol Version 4, Src: 10.243.33.126, Dst: 10.243.33.46
```

```
Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) Dual Band Wireless-AC 7265
   Physical Address. . . . . . . . . : 48-45-20-FA-27-B0
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   IPv6 Address. . . . . . . . . . . : 2404:c0:317f:694:fb76:535b:80be:8d5b(Preferred)
   Temporary IPv6 Address. . . . . . : 2404:c0:317f:694:a8b4:5c1f:e40:ea45(Preferred)
   Link-local IPv6 Address . . . . . : fe80::d64:472d:9285:d7cc%6(Preferred)
   IPv4 Address. . . . . . . . . . . : 10.243.33.126(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Tuesday, 31 March 2026 15:34:11
   Lease Expires . . . . . . . . . . : Tuesday, 31 March 2026 17:48:51
   Default Gateway . . . . . . . . . : fe80::805d:a3ff:fe62:d75a%6
                                       10.243.33.46
   DHCP Server . . . . . . . . . . . : 10.243.33.46
   DHCPv6 IAID . . . . . . . . . . . : 105399584
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2E-4C-01-72-60-45-CB-BD-63-AF
   DNS Servers . . . . . . . . . . . : 10.243.33.46
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

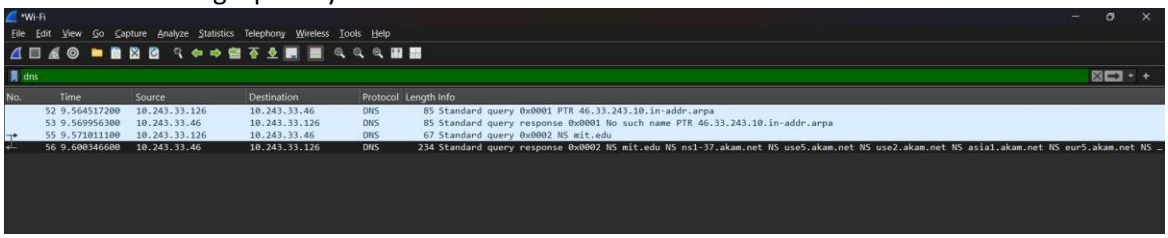
- Periksa pesan permintaan DNS. Apa ”jenis” atau ”type” dari pesan tersebut? Apakah pesan tersebut mengandung ”jawaban” atau ”answers”?
  Berdasarkan rincian pada *Domain Name System (query)*, jenis atau *type* dari pesan permintaan DNS tersebut adalah **Type NS** (*Name Server*). Hal ini menunjukkan bahwa sistem sedang mencari tahu *server* nama yang otoritatif untuk domain mit.edu

```
▼ Domain Name System (query)
    Transaction ID: 0x0002
  ▶ Flags: 0x0100 Standard query
    Questions: 1
    Answer RRs: 0
    Authority RRs: 0
    Additional RRs: 0
  ▼ Queries
    ▶ mit.edu: type NS, class IN
    [Response In: 56]
```

- Periksa pesan balasan DNS. Apa nama server MIT yang diberikan oleh pesan balasan? Apakah pesan balasan ini juga memberikan alamat IP untuk server MIT tersebut
  Terdapat **8 (delapan)** *name server* yang diberikan sebagai jawaban (**Answer RRs: 8**) Yaitu:
  ns1-37.akam.net
  use5.akam.net
  use2.akam.net
  asia1.akam.net
  eur5.akam.net
  asia2.akam.net
  usw2.akam.net
  ns1-173.akam.net

```
▼ Domain Name System (response)
    Transaction ID: 0x0002
  ▶ Flags: 0x8180 Standard query response, No error
    Questions: 1
    Answer RRs: 8
    Authority RRs: 0
    Additional RRs: 0
  ▼ Queries
    ▶ mit.edu: type NS, class IN
  ▼ Answers
    ▶ mit.edu: type NS, class IN, ns ns1-37.akam.net
    ▶ mit.edu: type NS, class IN, ns use5.akam.net
    ▶ mit.edu: type NS, class IN, ns use2.akam.net
    ▶ mit.edu: type NS, class IN, ns asia1.akam.net
    ▶ mit.edu: type NS, class IN, ns eur5.akam.net
    ▶ mit.edu: type NS, class IN, ns asia2.akam.net
    ▶ mit.edu: type NS, class IN, ns usw2.akam.net
    ▶ mit.edu: type NS, class IN, ns ns1-173.akam.net
    [Request In: 55]
    [Time: 29.335500 milliseconds]
```

- Sertakan hasil tangkapan layer



nslookup www.aiit.or.kr bitsy.mit.edu

```
C:\Users\ASUS ROG STRIX>nslookup www.aiit.or.kr bitsy.mit.edu
DNS request timed out.
    timeout was 2 seconds.
Server:  UnKnown
Address:  64:ff9b::1200:4803

DNS request timed out.
    timeout was 2 seconds.
DNS request timed out.
    timeout was 2 seconds.
DNS request timed out.
    timeout was 2 seconds.
DNS request timed out.
    timeout was 2 seconds.
*** Request to UnKnown timed-out

C:\Users\ASUS ROG STRIX>
```

karena tidak bisa maka saya pake

dns-ethereal-trace-4	03/09/2004 08:14	Wireshark capture ...	14 KB

- Ke alamat IP manakah pesan permintaan DNS dikirimkan? Apakah alamat IP tersebut merupakan default alamat IP server DNS lokal Anda?

```
▾ Internet Protocol Version 4, Src: 128.238.38.160, Dst: 18.72.0.3
```

Pesan permintaan DNS dikirimkan ke alamat IP **18.72.0.3**

Bukan. Alamat IP tersebut bukan merupakan DNS lokal default saya. Alamat **18.72.0.3** adalah alamat IP milik server **bitsy.mit.edu**

- Periksa pesan permintaan DNS. Apa ”jenis” atau ”type” dari pesan tersebut? Apakah pesan tersebut mengandung ”jawaban” atau ”answers”?

Jenis pesan tersebut adalah **Type A**

```
▾ Domain Name System (query)
    Transaction ID: 0x0003
  ▸ Flags: 0x0100 Standard query
    Questions: 1
    Answer RRs: 0
    Authority RRs: 0
    Additional RRs: 0
  ▸ Queries
    [Response In: 105]
```

- . Periksa pesan balasan DNS. Berapa banyak ”jawaban” atau “answers” yang terdapat di dalamnya. Apa saja isi yang terkandung dalam setiap jawaban tersebut?

Answer RRs: 1

Isinya adalah alamat IPv4 untuk host www.aiit.or.kr, yaitu **218.36.94.200**

```
▾ Domain Name System (response)
    Transaction ID: 0x0003
  ▸ Flags: 0x8180 Standard query response, No error
    Questions: 1
    Answer RRs: 1
    Authority RRs: 2
    Additional RRs: 2
  ▸ Queries
  ▾ Answers
    ▸ www.aiit.or.kr: type A, class IN, addr 218.36.94.200
  ▸ Authoritative nameservers
  ▸ Additional records
    [Request In: 104]
    [Time: 14.342000 milliseconds]
```

dns-ethereal-trace-4.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

dns

No.	Time	Source	Destination	Protocol	Length	Info
100	4.265296	128.238.38.160	18.72.0.3	DNS	82	Standard query 0x0001 PTR 3.0.72.18.in-addr.arpa
101	4.278516	18.72.0.3	128.238.38.160	DNS	212	Standard query response 0x0001 PTR 3.0.72.18.in-addr.arpa PTR BITSY.MIT.EDU NS W20NS.MIT.EDU NS BITSY.MIT.EDU NS STRAWB.MIT.EDU
102	4.279430	128.238.38.160	18.72.0.3	DNS	83	Standard query 0x0002 A www.aiit.or.kr.poly.edu
103	4.293283	18.72.0.3	128.238.38.160	DNS	135	Standard query response 0x0002 No such name A www.aiit.or.kr.poly.edu SOA gatekeeper.poly.edu
104	4.293517	128.238.38.160	18.72.0.3	DNS	74	Standard query 0x0003 A www.aiit.or.kr
105	4.307859	18.72.0.3	128.238.38.160	DNS	156	Standard query response 0x0003 A www.aiit.or.kr A 218.36.94.200 NS ns.aiit.or.kr NS w3.aiit.or.kr A 222.106.36.66 A 222.106.36.